Available Online: 5 June 2026 DOI: 10.54254/2977-3903/2026.34221

Game technology and artificial intelligence

Zixia Fan[1*], Yanyi Liu[2]

[1]School of Mathematics and Statistics, Henan Normal University, Xinxiang, China
[2]School of Mathematics and Statistics, Xinxiang University, Xinxiang, China

*Corresponding Author. Email: zhangsan00102@outlook.com

Abstract. Starting from the perspective of game players, this paper explores the technological issues behind online games and discovers the homology between game experience and scientific research. The legitimization of the game industry and the wave of artificial intelligence demonstrate the in-depth coupling of game technology and artificial intelligence, which is best exemplified by *Exploring the Path of AI Innovation: Report on the Innovative Development of Game Technology and Artificial Intelligence* issued by the Research Center for Virtual Economy and Data Sciences of the Chinese Academy of Sciences and the Key Laboratory of Big Data Mining and Knowledge Management of the Chinese Academy of Sciences. The comparative study of game technology and artificial intelligence between China and the United States is the innovation of this paper.

Keywords: game technology, artificial intelligence, homology, coupling

1. Introduction

In 2025, Fan Zixia, as a player of DecFovel from Henan Normal University, participated in the Solo Finals of the *League of Legends* National College Mobile Game Tournament and won the runner-up. The live broadcast of the event by Tencent Video in Guangzhou, and online media such as Beijing Weimeng Chuangke's Weibo accounts "Game Omniscient" and "Uncle You" all reported this incident, creating a certain network effect. According to DeepSeek Dialogue, player DecFovel of Henan Normal University is a strength-based competitor. However, as a contemporary college student, it is far from enough to only stay at the level of mobile game entertainment, at most becoming an e-sports player or a senior player. Paying attention to the supporting role of game technology in the game industry, especially the new revolution of artificial intelligence in the field of game technology, is of great significance for the "going global" development of China's game industry and the dissemination of Chinese culture [1].

2. The infancy of video games

The "infancy" of video games (1950s–1970s) was the initial collision stage between computer technology and entertainment demand. The technological experiments and creative buds in this period laid the foundation for the modern game industry. The infancy of games went through three transformations: the technological prototype stage, the sprouting of commercial attempts, and the pre-arcade era. 1. Technological prototype

Copyright: © 2026 by the authors. Submitted for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

stage (1950s): The core carrier was laboratory computers. In 1952, *OXO* (Tic-Tac-Toe), the first graphic display game of the University of Cambridge, ran on the EDSAC computer, spread only within academic circles with no possibility of commercialization. In 1958, *Tennis for Two* of Brookhaven National Laboratory, a simulated tennis game made by William Higinbotham with an oscilloscope, was first displayed to the public on the laboratory open day. The technical limitations of this stage were reflected in the reliance on room-sized vacuum tube computers, with outputs only being oscilloscope light spots or ASCII symbols. 2. Sprouting of commercial attempts (1960s): The breakthrough was moving from laboratories to public places. In 1962, *Spacewar!* of MIT, the first work "with modern game elements", was a two-player space shooting game spread to universities across the United States via the PDP-1 computer. In 1966, Ralph Baer put forward the concept of "TV game console", and the patent drawings included advanced designs such as light guns and multiplayer interaction. The industrial characteristics of this stage were that computers were costly, about 200,000 US dollars per unit, only for institutional use, and most developers were physicists or engineers rather than full-time game designers. 3. Pre-arcade era (1970–1972): The key innovation was the emergence of dedicated hardware. In 1971, *Computer Space*, the first commercial video game developed by Nutting Associates of the United States, was a simplified version of *Spacewar!* and failed commercially due to complex operation, with only 1,500 units sold. In 1972, the prototype of Atari's *Pong* was made by Alcorn with parts costing less than 500 US dollars, and business opportunities were found when the machine was stuffed with coins from players during testing. The technical turning point of this stage was reflected in two aspects: first, integrated circuits reduced hardware costs; second, the shift from general-purpose computers to dedicated game devices [2]. The analysis of game characteristics in infancy is shown in Table 1.

Table 1. Analysis of game characteristics in infancy

Dimension	Typical Performance	Modern Impact
Interaction mode	Physical buttons/knobs	Laid the basic layout of gamepads
Visual presentation	Monochrome vector graphics / oscilloscope light spots	Revival of minimalist aesthetics (*Geometry Dash*)
Content design	Abstract rules (no narrative)	Independent games still continue this tradition
Development mode	1–2 person amateur projects	Source of independent development culture

In the infancy of games, there were three forgotten founders. Thomas Goldsmith applied for the first electronic game patent, a cathode-ray tube entertainment device, in 1947. Steve Russell was the main creator of *Spacewar!* and later joined early game companies in Silicon Valley. Bill Pitts developed the first commercial arcade game *Galaxy Game* in 1971, with only 2 units produced. The historical enlightenment brought by the infancy of games is reflected in three aspects: first, technology drives creativity; limited by hardware performance, early designers were forced to explore fun in minimalism. Second, failure is a seed; the failure of *Computer Space* directly gave birth to the success of *Pong*. Finally, moving from niche to mass, from laboratory curios to arcades, defines the essence of video games: low-threshold interactive joy.

3. The legitimization of the game industry

On November 10, 2022, the European Parliament adopted a landmark resolution, formally calling on the European Union (EU) to formulate a long-term strategy to support the development of the video game

industry. This is the first time that EU institutions have explicitly recognized the cultural and economic value of the game industry and incorporated it into the policy framework. The main contents are as follows: 1. Core objectives of the resolution. Recognize games as an important part of the "cultural and creative industries", emphasize their cultural, educational and technological innovation value, require the European Commission to formulate a "European Video Game Strategy", coordinate the policies of member states, and provide financial support for game development through existing programs such as "Creative Europe". 2. Recognition of economic and cultural value. In 2021, the annual operating revenue of the European game industry reached 23.3 billion euros, accounting for 1/5 of the global market, and directly created at least 98,000 high-quality jobs. Emphasize the cross-border application potential of game technology in medical, education and other fields. 3. Key policy directions. First, talent training: set up special education programs to solve skill shortages. Second, industrial support: provide development funds and tax incentives for small and medium-sized studios. Third, technological research and development: support the development of underlying technologies such as game engines and VR/AR. Finally, cultural protection: spread European cultural heritage through game media. 4. Controversies and compromises. Some members of parliament proposed a "game addiction prevention and control" clause, which was ultimately not included in the main text of the resolution. The issue of micro-transaction regulation was left for discussion in the subsequent Digital Markets Act. 5. Follow-up impact. The European Commission must submit a policy response by June 2023, which may affect the allocation of the EU's "Creative Europe" budget for 2023–2027, and provide EU-level coordination for countries with existing game support policies such as France and Germany [3].

The resolution was adopted by an overwhelming majority of 560 votes in favor and 34 votes against, reflecting Europe's shift in the strategic positioning of the digital content industry. The European Game Developers Federation (EGDF) called this a "turning point for the European game industry", and the cultural circle pays attention to its synergistic effect on traditional art forms. Although the resolution has no direct legal effect, it has laid the foundation for subsequent legislation. EU game companies are expected to obtain a more unified cross-border development framework, especially emerging game development regions in Eastern Europe may benefit significantly.

4. Four milestones

Game technology plays a key role in the development of artificial intelligence, especially making significant contributions in four aspects: algorithm training, computing power breakthrough, human-computer interaction and scenario verification [4]. The specific role of game technology is analyzed for the four milestone stages of AI development as follows. 1. Early rule system stage (1950s–1980s): Representative technologies are chess AI and early NPC behavior trees. The contributions of game technology are: first, verifying symbolic AI. Before IBM's "Deep Blue" defeated the chess champion, game chess programs, such as Christopher Strachey's checkers program in 1951, verified the feasibility of rule systems for the first time. Second, behavior tree prototype. The computerization of *Dungeons & Dragons* in 1974 gave birth to the first NPC decision tree, which became the basis of modern game AI behavior models. Third, application of finite state machines. The ghost AI in the 1980 arcade game *Pac-Man* is still a teaching case. 2. The rise of machine learning (1990s–2010s): Representative breakthroughs are deep learning and reinforcement learning. The role of game technology is: first, supply of training environments. Since 2000, the *Doom* AI Competition has promoted the early development of reinforcement learning. Game physics engines (such as Havok) provide low-cost test sites for robot simulation. Second, algorithm acceleration verification. NVIDIA GPUs were originally designed for games and later became the foundation of deep learning computing power (CUDA architecture).

DeepMind verified the DQN algorithm in Atari games such as *Space Invaders* in 2013. 3. The explosive stage of deep learning (2012–2018): Key events are AlphaGo and Generative Adversarial Networks (GAN). The empowerment of game technology is: (1) Complex decision-making training ground. *StarCraft II* became a research platform for multi-agent collaboration, the predecessor of AlphaStar in 2017. OpenAI verified large-scale reinforcement learning in *Dota 2* in the 5v5 competition in May 2018. (2) Data generation tool. Game engines (Unreal, Unity) automatically generate annotated data sets to solve the shortage of computer vision data. GAN technology first attracted public attention through the *Pokémon* generation experiment. 4. The stage of large models and virtual-real integration (2019–present): Cutting-edge fields are multimodal large models and digital twins. The integration of game technology is: (1) Virtual world construction. *Microsoft Flight Simulator* uses AI to generate global 3D maps to train autonomous driving systems. The NVIDIA Omniverse platform realizes industrial digital twins relying on game engines. (2) Human-computer interaction revolution. Game motion capture technology (such as MetaHuman in *Fortnite*) accelerates the development of emotional AI. The game UI design paradigm affects the interaction design of dialogue systems such as ChatGPT. The supplementary special survey data is shown in Table 2.

Table 2. The supplementary special survey data

Technical Field	Proportion of Game Contributions (Research Sample)	Typical Cases
Reinforcement Learning	61%	OpenAI *Dota 2* AI
Computer Vision	38%	*Grand Theft Auto V* Autonomous Driving Dataset
Multi-Agent System	73%	*StarCraft* AI Competition
Physical Simulation	89%	NVIDIA Flex Liquid Simulation

The enlightenment from the four milestones is: (1) Low-cost trial and error. The game world provides a safe and reproducible AI training environment, for example, Waymo uses *Grand Theft Auto* to test extreme scenarios. (2) Technology feedback cycle. Game graphics technologies such as ray tracing are promoting the development of Neural Radiance Fields (NeRF). (3) New ethical challenges. Game social AI (such as *Replika*) triggers discussions on AI emotional manipulation. Facts have proved that game technology has become the "invisible infrastructure" for AI development, and this synergy may be further deepened in fields such as metaverse and brain-computer interface in the future. The *2023 AI Index Report* of Stanford University pointed out that about 32% of AI papers cite game-related experimental data, and this proportion is still rising.

5. Science and games

The interactive relationship between science and games is a dynamic process of two-way empowerment: science provides technical foundation and theoretical support for games, and games become a testing ground and communication carrier for scientific research [5]. This symbiotic relationship shows a clear evolutionary trajectory in the history of technological development, and we analyze it from four dimensions respectively. 1. Science as an incubator for games. (1) Supply of basic technologies. Computer science: Turing Machine Theory (1936) → programmable game logic. Mathematics: John Conway's *Game of Life* (1970) → cellular automata influencing simulation games. Physics: Rigid Body Dynamics (1980s) → physics engine of *Angry Birds*. (2) Key breakthrough cases. In 1962, *Spacewar!* was born at MIT Lincoln Laboratory, directly using the oscilloscope display technology of the PDP-1 computer. In 1996, *Quake* first applied the 3D rendering

algorithm developed by Carmack (later derived into the ID Tech engine). (3) Contemporary frontier. Quantum computing: The IBM Quantum Experience platform has launched the first batch of quantum games, such as *Quantum Labyrinth*. Neuroscience: In 2023, clinical trial of the brain-computer interface game *MindLabyrinth* for treating ADHD. 2. Games as an accelerator of science. (1) Innovation of scientific research paradigms. (See Table 3) (2) Technology verification platform. Autonomous driving: Waymo uses *Grand Theft Auto V* to simulate extreme traffic scenarios. Materials science: In 2022, the University of Cambridge modeled molecular structures in *Minecraft*. (3) Revolution in science communication. *Kerbal Space Program* is used by NASA as an aerospace education tool. *Plague Inc.* saw a 3,000% increase in daily active users during COVID-19, triggering epidemiological discussions. 3. Co-evolution model. Case of technological spiral upward: History of GPU development: In 1999, GeForce 256 was designed for games; in 2006, the CUDA architecture shifted GPUs to general-purpose computing; in 2023, the H100 chip trains AI models and renders *Cyberpunk 2077* simultaneously. 4. Future integration directions. (1) Gamification of scientific research. CERN of Switzerland developed *Particle Adventure* to explain the Higgs boson, and Stanford University uses VR games to train surgeons (surgical error rate reduced by 32%). (2) Scientizaion of games. The boss battles in *Elden Ring* have become samples for psychological stress response studies, and the tire physics model of *Forza Motorsport* is adopted by F1 teams. (3) Ethical challenges. Moral boundaries of research on game addiction mechanism and dopamine secretion, and whether metaverse virtual experiments need to follow the *Declaration of Helsinki*. The comparison between game solutions and traditional methods is shown in Table 3.

Table 3. The comparison between game solutions and traditional methods

Field	Game Solution	Comparison with Traditional Methods
Protein Folding	*Foldit* players solved a 15-year problem in 3 years	Pure algorithm calculation is time-consuming and power-consuming
Astronomy	*EVE Online* players classify galaxies	Professional teams are 50 times less efficient
Medicine	*Sea Hero Quest* detects dementia	Clinical testing costs reduced by 90%

China and the United States have different attitudes towards science and games. The United States emphasizes the transformation of basic science, such as the special "Games and Science" project set up by the US National Science Foundation (NSF). China focuses on technological applications, and China's "National Key R&D Program of China on Game Technology" includes brain science projects. Science historian Jon Peterson pointed out: "Video games are the most successful applied science product of the 20th century, but also the most underestimated basic scientific research tool." This dual attribute continues to release value in cutting-edge fields such as artificial intelligence and complex systems. The boundary between games and science may eventually completely dissolve like the "Oasis" world in *Ready Player One*.

6. The US NSF "Games and Science Initiatives"

The "Games and Science Initiatives" established by the US National Science Foundation (NSF) is a strategic project officially promoting the in-depth integration of game technology and scientific research, aiming to explore the unique value of games in scientific exploration, education communication and technological innovation. The core content and typical cases of this initiative are analyzed as follows. 1. Background and objectives of the initiative. (1) Establishment time: Launched a pilot in 2011 and upgraded to a permanent project in 2015. (2) Funding scale: Annual budget of about 12 million US dollars (2023 data). (3) Three missions: Transform game mechanics into scientific research tools, improve STEM education effects through

games, and explore the empowering potential of game technology for scientific discovery. 2. Key funding directions. (1) Gamification of scientific research crowdsourcing. Typical cases: *Eyewire* (MIT neuroscience project). Players help map the 3D atlas of retinal neurons through games, and the results were published in *Nature* (2016), discovering 6 new neuron types. *NOVA Labs* (PBS cooperative project), where players participate in real scientific research tasks such as solar storm prediction and RNA folding. (2) Development of scientific simulation engines. Breakthrough project: *Molecular Playground* of the University of Illinois, a quantum chemistry simulation platform based on the Unity engine, which can visualize electron cloud orbitals in real time (accuracy up to DFT level). *Climate Warriors* funded by NSF, a multi-agent climate model prediction game, cited by the IPCC as a model for public participation. (3) Innovation in educational methodology. Representative achievements: *CodeCombat*, a programming learning game, which improved high school students' Python learning efficiency by 40% (NSF Evaluation Report). *Cellverse*, a VR game, allowing middle school students to understand the mechanism of mitochondrial lesions through games. 3. Technology transformation path (See Table 4). The successful cases of scientific tools in game technology are shown in Table 4. 4. Characteristics of operation mechanism. (1) Cross-review system. The review committee includes game designers (such as former Blizzard chief designer Rob Pardo) and Nobel laureates (such as physicist Carl Wieman). (2) Industrial collaboration model. Projects are required to be jointly applied by universities (such as UC Berkeley Game Science Center) and game companies (such as Zynga). (3) Mandatory open source clause. All results must be published on GitHub, and 23 projects have become Apache incubator projects. 5. Controversies and challenges. (1) Effectiveness questioning. Some members of Congress criticized the scientific research contribution of projects such as *Foldit* as "difficult to quantify" (2021 Hearing Records). (2) Ethical risks. *NeurOlympics*, a brain science game, was suspended in 2022 due to potential epilepsy induction. (3) Industrial siphon effect. 30% of funded researchers eventually entered the game industry (NSF 2022 Talent Drain Report). Future layout: According to the NSF 2023–2028 Strategic Plan, it will focus on supporting quantum games, neurogames and metaverse scientific research. Quantum games: Develop game interfaces for quantum algorithm education. Neurogames: EEG-based attention training system. Metaverse scientific research: Establish virtual laboratory collaboration standards (cooperative project with Meta). The technology transformation path is shown in Table 4, and the China-US comparison is shown in Table 5.

Table 4. The technology transformation path

Game Technology	Scientific Application Field	Transformation Case
Voxel Rendering Engine	Geological Modeling	USGS Earthquake Simulation System
Multiplayer Online Architecture	Epidemic Spread Research	COVID-19 Supercomputer Simulation
Procedural Generation Algorithm	Exoplanet Atmosphere Simulation	NASA Exoplanet Database Construction

Table 5. China-US comparison

Dimension	US NSF Model	China's Corresponding Measures
Funding Focus	Development of Basic Scientific Research Tools	Industrial Technology Tackling (e.g., VR Medical Treatment)
Participants	Universities + Small and Medium-sized Enterprises	Led by Major Companies such as Tencent/NetEase
Typical Achievements	*Foldit* Citizen Science	*Honor of Kings* AI Juewu Laboratory

NSF Director Sethuraman Panchanathan said: "Games are reshaping the way of scientific discovery – from protein folding to galaxy evolution, we are witnessing a new revolution in scientific methodology [6]." The evolution of this initiative vividly interprets how "Serious Games" have grown from marginal innovation to national-level scientific research infrastructure.

7. Exploring the path of AI innovation: report on the innovative development of game technology and artificial intelligence

In July 2023, the Game Publishing Committee of the China Audio-Video and Digital Publishing Association and the China Game Industry Research Institute jointly released *Exploring the Path of AI Innovation: Report on the Innovative Development of Game Technology and Artificial Intelligence*. This report is a scientific research achievement researched and written by the Research Center for Virtual Economy and Data Sciences of the Chinese Academy of Sciences and the Key Laboratory of Big Data Mining and Knowledge Management of the Chinese Academy of Sciences. The abstract of this report states:

"With *AlphaGo* defeating human Go champions and OpenAI launching the ChatGPT large model, artificial intelligence is becoming one of the most far-reaching technological fields for human society in the 21st century. Scientific exploration, game technology and industrial demand are important driving forces for the breakthrough development of artificial intelligence, among which game technology is a relatively active factor. The report proposes that game technology is a comprehensive ecology of science, technology and humanities formed by the comprehensive application of advanced digital technology of the 21st century to the game industry. Specifically, game technology is a new technology cluster originating from the game industry and gradually improving with the development of the game industry, which provides increasing support for human needs such as leisure and entertainment, education and learning, industrial innovation, cultural communication and scientific exploration, and continuously builds a comprehensive ecology of game-related theories, technologies, ethics and culture in the process of continuous integration with computer, internet and digital technologies [7]."

The report systematically analyzes the root causes, course and future trends of game technology driving the innovative development of artificial intelligence. The report proposes that science and games are closely related; in the birth and development of science, many scientists revealed the mysteries and truths of nature by regarding science as an advanced game, so the view that games are an important driving force for scientific exploration has become the theoretical basis of this report. Artificial intelligence is a branch of computer science with strong scientific exploration attributes. Based on the common ground of computers, game technology has become an important driving force for the development of artificial intelligence, which is essentially an extension and example of games promoting science. Over the past 70 years, game technology and scientific exploration have formed a "double helix structure", jointly promoting the development of artificial intelligence. As empirical evidence, the report conducts an in-depth analysis of four milestone events: the origin and birth of artificial intelligence, the defeat of the world chess champion by IBM's Deep Blue supercomputer, the defeat of the world Go champion by the *AlphaGo* system, and the launch of the large model tool ChatGPT by OpenAI, and proposes the important role played by game technology behind them.

According to the calculation of the research cooperation team, for every 1% increase in the scale of the game industry, the total operating income of listed artificial intelligence enterprises increases by an average of about 142 million yuan. In the future, the driving scale of the game industry to the artificial intelligence industry will continue to increase, and it is expected to rise from 31.576 billion yuan in 2023 to 103.81 billion yuan in 2030, with an average annual growth rate of about 16%. Artificial intelligence has become the main

field of scientific and technological innovation in the world. A large number of leading IT companies such as Microsoft, DeepMind, Tencent, Sony, Google and OpenAI have continuously increased the integration of game technology and artificial intelligence to enhance their competitiveness, and have produced many achievements in science, industry, agriculture, finance, medical treatment, cultural tourism, smart cities and other fields. With the promotion effect of game technology on artificial intelligence being concerned and valued by countries around the world, many countries have successively launched relevant policies to promote the development of game technology.

The study found that countries with developed game markets are often also countries with strong artificial intelligence innovation capabilities. Through in-depth research on the course, laws, cases, industries and international competition of game technology driving the development of artificial intelligence, this report provides a useful reference for government departments, industry, academia, the public and other parties to better understand and apply game technology, and provides inspiration for better promoting the innovative development of artificial intelligence through game technology. At the same time, against the background of China's vigorous advocacy of basic theoretical scientific innovation, this report attempts to find answers to important issues such as how to understand the essence of science and what methods to carry out scientific research and exploration in the discussion of the relationship between games and science, game technology and artificial intelligence, which is also the important information conveyed by this report to society. This report expounds the homology and coupling between game technology and artificial intelligence. Future research in related fields will pay more attention to the ethical dimension of game technology and artificial intelligence, highlighting the in-depth integration of game technology and humanistic reflection in the era of artificial intelligence.

8. Conclusion

Research has proved that the immersive experience of game players is inseparable from the support of game technology. Artificial intelligence promotes the upgrading and transformation of game technology, injects new momentum into the development of China's game industry, and provides new technological support for the overseas development of China's game industry and the dissemination of Chinese culture [8]. The homology and coupling between games and scientific research, artificial intelligence and game technology provide a new perspective for us to understand the development of the game industry and artificial intelligence technology. Ethical research and humanistic reflection on game technology and artificial intelligence can provide a more robust guarantee for the future development of the game industry and artificial intelligence.

References

[1] Chapelle, C. A. (2025). Generative AI as game changer: Implications for language education. *System, 132*, Article 103672.

[2] Zhao, X. Y. (2018). Ideas on the development of computer games in China. *Academy*, (30), 12–13.

[3] People's Network. (2022). *European Parliament adopts first resolution on the development of video game industry*. https: //world.people.com.cn/n1/2022/1113/c1002-32565002.html

[4] Ye, Q. (2023, May 9). Can playing games promote the development of artificial intelligence? *Science and Technology Daily*, pp. 1–2.

[5] Wu, Y. S. (2004). Science and games. *Study Times*, (2), 1.

[6] Ploug, R., Rimer, E., Petersen, A. K. S., Scirea, M., & Brown, J. A. (2025). Evolving camouflages: A user-centric AI approach for game aesthetics. *International Journal of Computer Games Technology*, Article

3609613. https: //doi.org/10.1155/ijcg/3609613

[7] Research Center for Virtual Economy and Data Sciences, Chinese Academy of Sciences. (2023, July). *Exploring the path of AI innovation: Report on the innovative development of game technology and artificial intelligence*. http: //feds.ac.cn/index.php/zh-cn/2020-04-13-04-45-08/2944-ai

[8] Chang, J. H., Li, H., & Lu, T. (2025). AI empowerment: Enhancing the influence of Bay Area e-sports and sports exchanges. In *Proceedings of the 15th National Physical Training Conference* (Volume 2) (pp. 135–138). https: //doi.org/10.26914/c.cnkihy.2025.005534